MMF 1921 Exam May 13, 2024

Instructions: Closed book and closed notes except for one side of a 8.5 by 11 inch sheet of handwritten notes. Only a simple scientific non-financial calculator with no programming capability is allowed. Write **neatly** as this will aid in providing maximum partial credit. **Show All Work.**

IMPORTANT: Interpretation of the exam questions is part of the exam and so no questions will be taken DURING the exam that ask for clarification of the question. You MUST turn in this question sheet with your answer booklet or else your exam will NOT be marked.

A useful fact:

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and is invertible, then $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Problem 1 (8 points)

A warehouse can store 10000 units of a certain commodity. A commodities trader has access to the warehouse and currently has 2000 units of the commodity in storage in the warehouse. At the start of each month t, for the next 12 months ($t = 1, ..., 12$), the trader can buy an amount of the commodity at price p_t subject to the capacity of the warehouse at start of month t or can sell an amount of the commodity at price s_t at the start of the month subject to how much of the commodity is currently in the warehouse. In addition, there is a unit inventory cost i_t for holding the commodity in the warehouse which is incurred at the start of month t. The trader wishes to have no inventory of the commodity at the end of the year. Formulate an LP that maximizes profit from the buying or selling of the commodity over the 12 month time horizon.

Problem 2 (20 points)

Consider the (primal) LP

$$\begin{array}{ll} \text{minimize} & -4x_1 - 3x_2 - 2x_3 \\ \text{subject to} & 2x_1 + 3x_2 + 2x_3 \leq 6 \\ & -x_1 + x_2 + x_3 \leq 5 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \end{array}$$

(a) (3 points) Write the dual of this LP.

(b) (10 points) Solve the (primal) LP using the simplex method (you must use the version developed in class).

(c) (4 points) Use your work from (b) to show that at termination of the simplex method the dual problem is solved to optimality as well.

(d) (3 points) In the Simplex Method a basic feasible solution x^* is deemed optimal if the corresponding reduced costs r_N are non-negative. In particular, this condition was deduced from the following relation $c^T x - c^T x^* = r_N^T x_N \geq 0$

where $x_N \geq 0$. Note that this condition seems to be a *sufficient* condition since from the view point of just satisfying $r_N^T x_N \geq 0$ it might be that this could hold with some elements of r_N negative. Explain why $r_N \geq 0$ is a *necessary* condition for optimality.

Problem 3 (25 points)
Part 1
Consider the function $f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$.

(a) (3 points) Prove $x = \binom{1}{1}$ is the unique global minimizer of $f(x)$ over $\mathbb{R}^2$.

(b) (10 points) With an initial point $x^{(0)} = \binom{0}{0}$ apply two iterations of pure Newton's method for minimizing $f(x)$ over $\mathbb{R}^2$.

Part 2
Let $f(x) = x^2 - \frac{x^3}{3}$. Suppose that we apply a gradient-descent method (using a constant step length of 0.5 and initial point $x^{(0)} = 1$) for the problem of minimizing $f(x)$ subject to $x \in \mathbb{R}$.

(a) (3 points) Write out the formula for the $(k + 1)st$ iterate.

(b) (3 points) Derive a closed form expression for the kth iterate $x^{(k)}$.

(c) (3 points) Show that the sequence $\{x^{(k)}\}$ converges to a strict local min.

(d) (3 points) Find the rate of convergence of the sequence $\{x^{(k)}\}$.

Problem 4 (14 points)

Consider the following variant of mean-variance portfolio optimization for constructing a porfolio of n assets

where one maximizes risk adjusted return subject to only a budget constraint with short-selling allowed i.e.

$$\begin{array}{ll} \min\limits_{x,y} & \mu^T x - \frac{1}{2}\gamma x^T Q x \\ \text{s.t.} & 1^T x = 1 \end{array}$$

where μ is the expected return vector, Q is the covariance matrix and $\gamma > 0$ is a risk-aversion coefficient.

(a) (7 points) Find a closed-form solution for the problem above. State the conditions under which the closed-form solution is well defined and unique and justify your answer.

(b) (3 points) Does the closed-form solution in (a) have any interpretation? Explain BRIEFLY why or why not.

(c) (4 points) Now consider the problem above but with no short selling allowed. Find the KKT conditions for this problem.

Problem 5 (8 points)

You are analyzing an investment opportunity with two stocks A and B with the expected return of Stock A $\mu_A = 0.025$, the expected return of Stock B $\mu_B = 0.015$, the variance of Stock A $\sigma_A^2 = 0.0049$, the variance of Stock B $\sigma_B^2 = 0.0025$, and $\sigma_{AB} = -0.001$.

Suppose that the expected returns were estimated using the last 20 monthly observations. Now consider the following robust optimization problem with an ellipsoidal uncertainty set

$$
\begin{array}{ll}
\min\limits_{x,y} & x^T Q x \\
\text{s.t.} & \mu^T x - y \geq 1\% \\
& y^2 \geq x^T \Theta x \\
& 1^T x = 1 \\
& y \geq 0
\end{array}
$$

Use this information to check whether the solution $x = [0.425\ 0.575]^T$ satisfies the KKT conditions of the problem above. (Note: Assume the target return constraint is active.)

Problem 6 (8 points, 4 points each)

Consider the following function

$$f(x) = \sqrt{x^T Q x} - \lambda \mu^T x$$

where $x \in \mathbb{R}^n$ is a vector of asset weights, $\mu \in \mathbb{R}^n$ is the vector of expected returns and $Q \in \mathbb{R}^{n \times n}$ is the PSD covariance matrix. Now consider the following optimization problem

$$
\begin{gathered}
\min_y \tfrac{1}{2} y^T Q y - \lambda \mu^T y - c \sum_{i=1}^n b_i \ln(y_i) \\
\text{s.t. } y \geq 0
\end{gathered}
$$

where $\lambda > 0, c > 0$ and $b_i > 0$ for $i = 1, ..., n$.

(a) Derive the individual risk contribution of each asset towards the function $f(x)$.

(b) Using the optimization model above, find an expression where, at optimality, relates the risk contribution of asset i to asset j. This expression should be in terms of y_i and y_j (you do not need to normalize your result).

Problem 7 (7 points)

(a) (2 points) Why optimize VaR or CVaR?

(b) (5 points) Define and explain the role of function $F_\alpha(x, \gamma)$? Be as detailed as you can.

Problem 8 (10 points)

Consider the problem

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array}$$

where $f(x)$ is some differentiable function from $\mathbb{R}^n$ to $\mathbb{R}^1$.

(a) (3 points) Formulate the barrier problem for the problem above and derive the KKT conditions

(b) (4 points) Using your results from (a) write out the system of equations that when solved would generate the search direction for a primal-dual interior point method for the problem above.

(c) (3 points) What are the requirements on the function $f(x)$ in order for the Newton system in (b) to be well-defined and a decent direction?